

HOMEWORK 5

Problem 1 Use the provided dataset “HW5_data.csv” for this question. Consider the linear model

$$y = X\beta + \sigma\epsilon$$

where y is a $n \times 1$ vector of observations, X is the $n \times p$ design matrix, β is the $p \times 1$ vector of coefficients, σ^2 is the variance and ϵ is a $n \times 1$ vector of independent standard normal random variables. For the data set, $n = 200$ and $p = 5$.

- (1) (8 pts) Using the notation in the slides, matrices Q , P and Λ are such that

$$QX'XQ' = \Lambda^2 \quad PXQ' = D = \begin{pmatrix} \Lambda \\ 0 \end{pmatrix}, \quad \Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_p)$$

where the (λ_j) are the square roots of the eigenvalues of $X'X$. Show that if $\nu = P\epsilon$ then $E(\nu_i) = 0$ and $\text{Var}(\nu_i) = 1$ for all $i = 1, \dots, n$.

- (2) (10 pts) If $\gamma = Q\beta$, show that $PX\beta$ can be written as $D\gamma$. Hence, show that the original linear model can be written as

$$z_i = \begin{cases} \lambda_i \gamma_i + \sigma \nu_i & i = 1, \dots, p \\ \sigma \nu_i & i = p + 1, \dots, n, \end{cases}$$

where $z = Py$.

- (3) (10 pts) Prove that

$$\sum_{j=1}^p \text{Var}(\hat{\beta}_j) = \sigma^2 \sum_{j=1}^p \frac{1}{\lambda_j^2}.$$

(Hint, the trace of a matrix is the sum of its eigenvalues.)

- (4) (10 pts) From the data set provided, find $\hat{\gamma}$.

- (5) (12 pts) Now suppose we want to keep m components where m is the smallest integer for which

$$\frac{\sum_{j=1}^m \lambda_j^2}{\sum_{j=1}^p \lambda_j^2} > 0.9.$$

Find m and the corresponding $\hat{\beta}_{pca}$

Problem 2

Consider the Bayesian linear model

$$y_i \sim N(x_i\beta, \sigma^2), \quad i = 1, \dots, n,$$

where $\sum_{i=1}^n x_i = 0$, $\sum_{i=1}^n x_i^2 = n$ and $\sum_{i=1}^n x_i y_i = \gamma$.

The prior for β is given by

$$\pi(\beta) = 0.5 \mathbf{1}(\beta = 0) + 0.5 N(\beta|0, \tau^2),$$

where $\mathbf{1}(\beta = 0)$ is the probability mass function which puts all probability on $\beta = 0$.

Suppose σ and τ are known.

(1) (8 pts) Write down the posterior density for β and note that this is always of the form prior times likelihood.

(2) (12 pts). By integrating out β , find the marginal likelihood

$$m(y_1, \dots, y_n) = \int \prod_{i=1}^n N(y_i | x_i \beta, \sigma^2) N(\beta | 0, \tau^2) d\beta.$$

(3) (12 pts) Hence, show that

$$\pi(\beta | y_1, \dots, y_n) \propto \left\{ \prod_{i=1}^n N(y_i | 0, \sigma^2) \right\} \mathbf{1}(\beta = 0) + m(y_1, \dots, y_n) N(\beta | \nu, \psi^2)$$

and find ν and ψ .

(4) (8 pts) Hence, write down $\pi(\beta = 0 | y_1, \dots, y_n)$.

(5) (10 pts) What is the significance of the term

$$\frac{m(y_1, \dots, y_n)}{\prod_{i=1}^n N(y_i | 0, \sigma^2)}$$

and show that this converges to infinity as γ^2 grows.